

BUSINESS CASE FOR EXECUTIVE APPROVAL

AI-Assisted KYC Digital Onboarding System

Northridge Bank of Canada (Fictional Case Study)

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Document Change Log

Version	Date	Author	Description
1.0	May 15, 2026	Namrata Modha	Initial draft for portfolio submission

1. Executive Summary

Recommendation: Approve the Full AI-Assisted KYC Digital Onboarding Platform. This is the only option that simultaneously resolves the Bank's regulatory exposure, operational cost inefficiency, and customer acquisition gap. Projected 3-year net value of CAD \$17.8M with an estimated payback period of under 4 months.
Supporting financial model: see Appendix A (KYC_BusinessCase_FinancialModel_Namrata_Modha.xlsx).

Northridge Bank of Canada's current Know Your Customer (KYC) onboarding process is manual, slow, and increasingly misaligned with Canadian regulatory expectations. FINTRAC strengthened photo identification requirements in December 2025, the 2025-2026 FATF mutual evaluation of Canada's AML regime is underway, and the launch of Consumer-Driven Banking in early 2026 is raising customer expectations for frictionless digital account opening.

The Bank currently onboards approximately 150,000 new customers annually at an estimated operational cost of CAD \$4.5M per year. A 60% false positive rate in AML alert screening consumes an additional CAD \$2.25M in compliance analyst hours annually. A 50% account application abandonment rate represents a material and quantifiable loss of prospective customers to competitors who offer faster digital onboarding experiences.

This Business Case evaluates three strategic options. The recommended option, full implementation of an AI-assisted KYC platform, will reduce per-verification cost by approximately 77%, cut false positive alerts by 55%, improve application completion rates, and position the Bank for full compliance with current and anticipated FINTRAC and FATF requirements. Total Year 1 investment is CAD \$2.35M. Annual net benefit from Year 2 is CAD \$6.99M.

2. Business Problem and Context

2.1 Background

KYC compliance is a legal obligation for all Canadian Schedule I banks under the Proceeds of Crime (Money Laundering) and Terrorist Financing Act (PCMLTFA), enforced by FINTRAC. Banks must verify customer identity at account opening, screen against sanctions and Politically Exposed Persons (PEP) lists, assess risk profiles, and maintain auditable records of all due diligence activities.

Northridge Bank's KYC process was designed for branch-based account opening. As digital banking has expanded, the same manual verification workflow has been retrofitted onto digital channels, producing a slow, inconsistent experience that fails both the customer and the compliance team.

2.2 Regulatory Drivers (Non-Negotiable)

- Mandatory photo identification for all account openings, profile updates, and high-risk transactions, effective December 15, 2025 (PCMLTFA amendment)
- FATF mutual evaluation of Canada's AML/ATF regime is in progress in 2025-2026; deficiencies found will trigger mandatory remediation obligations for all Canadian reporting entities
- Open Banking (Consumer-Driven Banking) framework launching early 2026 requires robust digital identity verification as the trust foundation for third-party data access
- Beneficial ownership tracing obligations as of October 1, 2025 require identification of all natural persons with 25%+ ownership for corporate accounts
- FINTRAC issued CAD \$25M+ in penalties across 23 notices of violation in 2024-2025, the highest in its history; TD Bank's USD \$3.09B AML settlement in 2024 demonstrated the scale of systemic exposure

2.3 Business Drivers (Strategic)

- Canadian fraud losses reached CAD \$643M in 2024, nearly 300% higher than 2020; only 5-10% of fraud incidents are reported, meaning actual losses are materially higher
- Synthetic identity fraud increased 153% in the first half of 2024; manual document review cannot reliably detect AI-generated synthetic identities
- 50% of new bank account applications are abandoned before completion; every percentage point of abandonment represents lost customer acquisition at scale
- 70% of Canadian financial institutions have already adopted AI-driven KYC verification; the Bank risks competitive disadvantage by maintaining a manual process

3. Current State Analysis (AS-IS)

3.1 Process Overview

A new customer initiates account opening through the Bank's digital portal or a branch. The compliance team manually reviews government-issued identification documents, cross-references the customer against sanctions lists and PEP registries using a legacy database tool, assigns a risk score based on analyst judgment, and either approves or escalates for Enhanced Due Diligence (EDD). The average time from application submission to account activation is 3 to 5 business days for digital applications.

3.2 Key Pain Points

Pain Point	Current Metric	Business Impact
High per-verification cost	~CAD \$30 per customer (manual review, analyst time, legacy tooling)	CAD \$4.5M/year for 150,000 onboardings
High AML false positive rate	~60% of screening alerts are false positives (industry range: 42-95%)	CAD \$2.25M/year in analyst time investigating non-suspicious activity
Application abandonment	50% of applications started are not completed	~150,000 lost prospective customers annually; direct revenue opportunity loss
Manual document verification gap	Human review cannot detect AI-generated synthetic IDs	Growing fraud exposure; FINTRAC audit risk; reputational risk
No straight-through processing	100% of applications require human touchpoint	Compliance team bottleneck; 3-5 business day activation delay
Photo ID compliance gap	Current process does not fully meet December 2025 PCMLTFA photo ID requirement at scale	Active regulatory exposure; potential FINTRAC notice of violation

3.3 Total Cost of Current State

Cost Category	Annual Estimate (CAD)
KYC operational cost (150,000 verifications x \$30)	\$4,500,000
False positive investigation cost (90,000 alerts x \$25)	\$2,250,000
Lost new account revenue (conservative estimate)	\$2,250,000
Regulatory risk exposure (potential FINTRAC penalties)	Up to \$20,000,000 per serious offense
Total Quantified Annual Cost	\$9,000,000

4. Strategic Options Analysis

Option 1: Status Quo (Do Nothing)

Maintain the existing manual KYC process with no material changes.

Annual cost: CAD \$6.75M+ in direct operational costs, excluding regulatory exposure.

Compliance posture: Active and growing exposure to FINTRAC violations. December 2025 photo ID requirements are not fully met at scale. FATF evaluation underway.

Customer experience: No improvement to the 50% abandonment rate. Continued loss of customers to competitors with faster digital onboarding.

Fraud risk: No capability improvement to detect synthetic identity fraud, which increased 153% in the first half of 2024.

Verdict: Not viable. Regulatory exposure alone makes this option unacceptable independent of cost and revenue considerations.

Option 2: Process Optimization Only (No AI)

Standardize the existing manual workflow, retrain compliance staff, introduce checklist-based tooling, and add a basic document upload portal for digital channels. No AI or machine learning components.

One-time investment: CAD \$250,000 (process redesign, training, basic portal).

Annual savings: ~15% efficiency gain, approximately CAD \$1M/year.

Compliance posture: Partial improvement. Manual document review still cannot detect synthetic identities. Photo ID requirement met inconsistently.

Abandonment rate: Marginal improvement. Fundamental friction of manual review and 3-5 day activation time remains.

False positive rate: No material improvement. AML screening still relies on legacy tool.

Verdict: Insufficient. Addresses operational efficiency marginally but does not resolve regulatory risk, fraud exposure, or competitive positioning. A short-term fix that defers the inevitable at additional cost.

Option 3: Full AI-Assisted KYC Platform Implementation (Recommended)

Implement an AI-powered KYC platform integrating automated document verification (OCR with liveness detection), biometric matching, real-time AML and sanctions screening, automated risk scoring, and straight-through processing for low-risk applicants. Platform deployed as SaaS via API integration with the Bank's existing core banking and CRM systems.

Capabilities: Document verification and biometric matching in under 35 seconds (FNZ benchmark). Straight-through processing for 78%+ of low-risk applicants with no human touchpoint required. Real-time sanctions and PEP screening. Automated beneficial ownership tracing for corporate accounts. Full audit trail for FINTRAC compliance.

Compliance posture: Full alignment with December 2025 PCMLTFA photo ID requirements. Built-in FINTRAC-compliant audit logging. Supports FATF mutual evaluation readiness. Open Banking identity verification foundation.

Fraud protection: AI-powered synthetic identity detection. 1-in-20 fraudulent verification attempts currently evade manual review; AI models trained on millions of document patterns reduce this materially.

Verdict: Recommended. Addresses all three dimensions: regulatory compliance, operational cost, and customer experience.

5. Cost-Benefit Analysis

5.1 Implementation Costs: AI-Assisted KYC Platform

Cost Item	Type	Year 1 (CAD)	Year 2-3 (CAD)
AI KYC Platform Licensing (SaaS)	Annual	\$600,000	\$600,000/yr
System Integration and API Development	One-time	\$800,000	\$0
Change Management and Staff Training	One-time	\$200,000	\$0
Data Migration, UAT, and Go-Live Support	One-time	\$150,000	\$0
Ongoing Maintenance and Support	Annual	\$100,000	\$100,000/yr
TOTAL		\$1,850,000	\$700,000/yr

5.2 Quantified Annual Benefits (from Year 1)

Benefit	Basis	Annual Value (CAD)
KYC operational cost reduction	\$30 to \$7 per verification x 150,000 customers	\$3,450,000
False positive alert reduction	55% reduction in 90,000 alerts x \$25 analyst cost each	\$1,237,500
Reduced abandonment revenue recovery	Abandonment improved from 50% to 30%; 15,000 net new accounts x \$150 avg first-year revenue	\$2,250,000
Regulatory penalty avoidance (risk-adjusted)	Conservative 5% probability of FINTRAC notice x \$15M avg penalty	\$750,000
TOTAL ANNUAL BENEFIT		\$7,687,500

5.3 Return on Investment Summary

Metric	Value
Total Year 1 Investment	CAD \$1,850,000
Total Year 1 Net Benefit	CAD \$5,837,500
Annual Net Benefit (Year 2+)	CAD \$6,987,500
Estimated Payback Period	Approximately 3.7 months
3-Year Net Present Value (8% discount rate)	CAD \$17,800,000
3-Year ROI	Approximately 860%

Note: All figures are based on stated assumptions in Section 11. Regulatory penalty avoidance is risk-adjusted and represents expected value only, not a guaranteed outcome. Revenue recovery estimates are conservative and do not account for customer lifetime value beyond Year 1.

6. Risk Assessment

ID	Risk	Likelihood	Impact	Mitigation	Owner
R-01	Vendor lock-in to single AI KYC platform	Medium	Medium	Select vendor with documented API exit strategy; negotiate data portability in contract; evaluate minimum 3 vendors (Onfido, Trulioo, Jumio) before award	Procurement
R-02	False negative rate in AI document verification allows fraudulent accounts	Low	High	Retain human review threshold for high-risk accounts; configure AI confidence score floor; monitor false negative rate monthly against SLA; quarterly model performance review with vendor	Compliance
R-03	PIPEDA compliance gap in biometric data handling	Medium	High	Legal review of vendor data processing agreements before contract signing; explicit consent capture for biometric matching; data residency confirmed to Canadian servers; Privacy Impact Assessment completed pre-launch	Legal / Privacy
R-04	Core banking system integration complexity delays go-live	Medium	Medium	API integration scoped and sized in discovery phase; integration testing environment provisioned 8 weeks before UAT; vendor integration engineer engaged throughout implementation	Technology
R-05	Staff resistance to change; compliance team displacement concerns	Medium	Low	Frame AI as analyst augmentation, not replacement; compliance team redeployed to higher-value EDD and investigations work; change management plan initiated at project kick-off	HR / Change Mgmt
R-06	Regulatory landscape shifts further before implementation completes	Low	Medium	FINTRAC regulatory monitoring assigned to Compliance team throughout project; vendor contract requires regulatory update SLA within 60 days of any PCMLTFA amendment	Compliance

7. Recommendation

This Business Case recommends approval of the Full AI-Assisted KYC Digital Onboarding Platform, with a requested Year 1 budget authorization of CAD \$1,850,000.

The status quo is not viable given active regulatory exposure under the December 2025 PCMLTFA amendments and the ongoing FATF evaluation. Process optimization alone is insufficient: a process-only intervention reduces cost marginally but does not address synthetic identity fraud, does not resolve the abandonment problem, and does not position the Bank for Open Banking identity requirements. Deferring a technology solution by 12-18 months to implement Option 2 first increases both regulatory risk and total cost of change.

The AI-Assisted KYC Platform is the only approach that simultaneously satisfies the regulatory obligation, addresses the operational cost inefficiency, improves customer acquisition, and creates a scalable identity infrastructure for Open Banking. The payback period of under 4 months and 3-year NPV of CAD \$17.8M make this a financially unambiguous decision. The primary risks are manageable through vendor selection discipline, legal review, and change management.

The Bank should move to vendor evaluation and procurement immediately. Given the active FATF evaluation and the December 2025 photo ID compliance gap, a target go-live within 9 months is both necessary and achievable.

8. High-Level Implementation Roadmap

Phase	Timeline	Key Activities	Milestone
Phase 1: Discovery and Procurement	Months 1-2	Requirements finalization; RFP issued to shortlisted vendors (Onfido, Trulioo, Jumio); vendor demos and scoring; legal review of data processing agreements; Privacy Impact Assessment initiated; procurement award	Vendor Selected
Phase 2: Integration and Build	Months 3-6	API integration with core banking and CRM systems; compliance team workflow redesign; staff training program developed and delivered; FINTRAC audit trail configuration and validation; test environment provisioned	Integration Complete
Phase 3: UAT and Pilot	Months 6-7	User Acceptance Testing with compliance team and pilot customer group; false positive and false negative rate benchmarked against SLA; performance testing under load; PIPEDA consent flow validated; go/no-go decision gate	UAT Sign-Off
Phase 4: Go-Live and Stabilization	Months 8-9	Phased rollout: digital channel first, then branch-assisted; hypercare support period; real-time monitoring of straight-through processing rate, false positive rate, and abandonment rate; compliance team transition to EDD-focused work	Go-Live
Phase 5: Optimization	Month 10+	Monthly KPI review against baseline; model tuning based on production data; vendor roadmap review for FATF and Open Banking capability additions; benefits realization report submitted to sponsors	Steady State

9. Success Metrics and KPIs

KPI	Current Baseline	Target (12 Months)	Measurement
Per-verification cost	CAD \$30	CAD \$7 or less	Monthly finance report
AML false positive rate	~60%	30% or less	Compliance dashboard
Account application abandonment rate	~50%	30% or less	Digital analytics platform
Straight-through processing rate	0% (all manual)	75%+	Platform SLA dashboard
Time to account activation (digital)	3-5 business days	Under 35 minutes	Core banking system
FINTRAC audit trail coverage	Partial (manual records)	100% automated logging	Compliance audit
FINTRAC notices of violation	Active exposure	Zero within 12 months of go-live	FINTRAC correspondence
Year 1 net benefit realization	N/A	CAD \$5.8M+	Benefits realization report

10. Key Assumptions and Dependencies

10.1 Assumptions

- Annual new customer onboarding volume of 150,000 is representative of the Bank's current run rate.
- Current manual per-verification cost of CAD \$30 includes direct analyst time, legacy tooling cost, and allocated overhead. Source: industry estimates (CAD \$25-50 range, conservative mid-point applied).
- AML false positive rate of 60% is within the published industry range of 42-95% for manual screening systems (KYC2020 research).
- Account application abandonment rate of 50% is per published Canadian banking research.
- AI platform reduces false positives by 55%, within the published 50-66% reduction range for AI-enhanced screening systems.
- Abandonment rate improvement from 50% to 30% is conservative; some vendors report improvements to 20% or below with optimized digital flows.
- Net new account revenue estimate of CAD \$150 per account in Year 1 is conservative and does not include lifetime value or cross-sell.
- Regulatory penalty risk-adjustment of 5% annual probability is an internal risk committee estimate, not a guaranteed outcome.
- Vendor pricing estimates are based on published SaaS pricing ranges for Onfido, Trulioo, and similar platforms; actual costs will be confirmed through RFP.

10.2 Dependencies

- Executive approval of budget authorization (CAD \$1,850,000 Year 1) before procurement can proceed.
- Legal and Privacy team availability to complete Privacy Impact Assessment and vendor data processing agreement review within Phase 1 timeline.
- Technology team resource allocation for API integration with core banking and CRM systems during Phase 2.
- FINTRAC regulatory environment remains substantially aligned with current PCMLTFA requirements through implementation period.
- Selected vendor maintains minimum 99.5% uptime SLA and commits to Canadian data residency for all biometric and identity data.

11. Regulatory and Compliance Reference

Regulation / Body	Relevance to This Business Case
PCMLTFA (Proceeds of Crime Money Laundering and Terrorist Financing Act)	Primary KYC legislative obligation for Canadian Schedule I banks. Amended December 2025 to require photo identification for account opening and high-risk transactions. Penalties up to CAD \$20M per serious offense.
FINTRAC	Enforces PCMLTFA. Issued CAD \$25M+ in penalties across 23 notices of violation in 2024-2025, the highest in its history. Publishes names of non-compliant entities. KYC audit trails must be producible on demand.
FATF (Financial Action Task Force)	International AML/ATF standard-setter. Conducting mutual evaluation of Canada in 2025-2026. Deficiencies identified will trigger mandatory remediation with defined timelines for all Canadian reporting entities.
Consumer-Driven Banking (Open Banking)	Framework launching early 2026. Requires robust digital identity verification as the trust foundation for consumer-authorized third-party data access. Banks without AI-assisted KYC will face structural gaps in Open Banking readiness.
PIPEDA	Governs collection, use, and storage of personal and biometric information. Biometric data requires explicit consent, purpose limitation, data minimization, and documented audits. Privacy Impact Assessment required before go-live.

Appendix A: Financial Model

The financial figures presented throughout this Business Case are supported by a dynamic Excel financial model attached as a separate workbook.

File: KYC_BusinessCase_FinancialModel_Namrata_Modha.xlsx

The workbook contains five sheets:

- Assumptions: all input variables with sources. Change any blue cell to run scenarios instantly across the full model.
- Current State: annual cost breakdown of the existing manual KYC process, fully formula-driven from the Assumptions sheet.
- Options Comparison: side-by-side financial comparison of all three strategic options with formula-linked outputs.
- Cost-Benefit Model: 3-year projection including year-by-year net cash flow, cumulative cash flow, NPV at 8% discount rate, ROI, and payback period.
- Sensitivity Analysis: what-if tables showing Year 1 net benefit and payback period under different volume and platform cost scenarios.

Colour coding follows industry standard: blue text denotes hardcoded inputs, black text denotes formulas, green text denotes cross-sheet references. No values are manually hardcoded outside the Assumptions sheet.

12. Executive Approval

By signing below, the approving parties confirm they have reviewed this Business Case and authorize the recommended course of action and associated budget as specified.

Name	Title	Date	Decision
Namrata Modha	Business Systems Analyst (Author)	May 15, 2026	Submitted
[Sponsor Name]	VP, Digital Banking		Pending
[Finance Name]	Chief Financial Officer		Pending
[Compliance Name]	Chief Compliance Officer		Pending
[Technology Name]	Chief Technology Officer		Pending

Northridge Bank of Canada is a fictional institution created for BSA portfolio purposes. All financial figures are estimates based on published industry research and are clearly cited within the document.